

Josiah Lo

📍 Concord, CA

✉ josiahlo@berkeley.edu

📞 (925) 566-4493

🌐 [linkedin.com/in/josiahdavidlo](https://www.linkedin.com/in/josiahdavidlo)

WORK EXPERIENCE

EMCORE Corporation Quartz MEMS Division — *Mechanical Engineer*

Jan 2026 - Present

- Created an opportunity for >\$1M in inertial unit sales by creating a Python script to analyze test history across ~1,000 scrapped accelerometers, identifying 276 units (~\$100K in parts) meeting a new customer's specifications; this script allowed my team to identify an additional 375 viable gyroscopes (~\$230K)
 - Confirmed root cause of accelerometer yield failures by pulling SQL testing data across 179 sensors, developing a MATLAB pipeline to isolate a discrete -15°C frequency anomaly in 50 units, showing via Ansys structural-modal Finite Element Analysis that moisture/frost contamination could reproduce the observed frequency shift, and sending dozens of units for mass spectroscopy testing to verify this hypothesis
 - Saved \$6,935 in parts cost and preserved \$147K in shipped IMU revenue by identifying units to undergo a reworking process (drill, vacuum-bake, reseal) that recovered 77% of moisture contaminated units
 - Designed and executed a 40 simulation Finite Element Analysis study to evaluate 8 die package designs of core accelerometer using coupled Thermal, Structural, Modal, and Vibration simulations in Ansys Mechanical to compare sensor performance and survivability—prompting the pilot production run of the best simulated design
-

EDUCATION

University of California, Berkeley

Expected Graduation May 2027

Bachelor of Science in Mechanical Engineering

GPA: 4.000

Dean's List - College of Engineering, All Semesters

Relevant Coursework: Behavior of Materials, Mechanics, Circuit Design, 3D Modeling, Manufacturing and GD&T, Finite Element Analysis, Feedback Control Systems

RELEVANT PROJECTS

UC Berkeley Solar Vehicle Team — *Project Lead*

Oct 2025 - Present

- Evaluated composite laminate suspension box for required safety factor using Ansys ACP FEA — Improved design to increase safety factor from 0.09 to 1.03 while limiting weight increase to 8.9%
- Redesigned roll cage in SolidWorks to optimize strength-to-weight ratio, validated via Ansys Structural FEA to verify safety factor and max deflection, and selected roll cage material to optimize for strength, cost, and manufacturability

6-Axis Surgical Robot — *Personal Project*

Dec 2025 - May 2026

- Designed the robotic arm in SolidWorks and fabricated the components with 3D printing, including a custom 2-stage 15:1 cycloidal gearbox at the base joint

- Routed and integrated wiring through the arm to maintain 6-axis motion clearance with 20-inch reach radius

Smart Window — *UC Berkeley Electronics Project*

Nov 2025 - Dec 2025

- Designed an autonomous smart window integrating motorized actuation, locking, and polymer-dispersed liquid crystal tint control, with temperature, smoke, and noise sensing to provide comfort, safety, and privacy automation
 - Designed custom control circuitry to interface three ESP32 microcontrollers with sensors, motors, and actuators, accounting for power distribution and signal requirements — programming them in Python to enable fully autonomous operation, including Wi-Fi communication, distributed sensor data sharing, and closed-loop control logic
 - Modeled and 3D-printed enclosures for electronics, optimizing for wiring clearance and component accessibility
-

SKILLS

Software and Programming: Ansys Mechanical, SolidWorks, Creo, AutoCAD, LTspice, MATLAB

Fabrication & Testing: 3D Printing, Mill, Lathe, GD&T per ASME Y14.5, Soldering, Lock-in Amplifier, PLL